A Derivation of Wien's Law by Dimensional Analysis

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Abstract

The Wien displacement law is the capstone of classical physics. – Wannier [1]

A derivation of Wien's law by dimensional analysis is presented. It suggests existence of a universal constant that relates frequency to energy allowing the construction of a dimensionless quantity from ω/T. Planck's constant is indeed such a constant.

Derivation

What we want to determine is the energy spectral density per unit volume, u_ω, of electromagnetic wave in empty space in equilibrium with surrounding matter at temperature T. We know that temperature is proportional to the average energy of the system and that Boltzmann constant k relates temperature to energy. Since we want to find the spectral density of electromagnetic wave, let us use the speed of light c [m/s] and its angular frequency ω [rad/s] as well as kT [J]. There is no other physical quantity to specify this system.

The Rayleigh–Jeans law is obtained by constructing the energy spectral density per unit volume [J/(rad/s)/m^3] from these three quantities:

$$u_\omega = A\,\frac{kT}{c^3}\,\omega^2, \tag{1}$$

where A is a dimensionless constant. The spectral density, u_ω, must satisfy Stefan's T^4 law:

$$u(T) = \int_0^\infty u_\omega\,d\omega = a\,T^4. \tag{2}$$

Suppose that this dimensionless constant A is in fact a function of ω and T. We can write such a function as $A(\omega T^r)$.[1] Inserting (1) into (2) yields

$$u(T) = \frac{kT}{c^3} \int_0^\infty \omega^2 A(\omega T^r)\, d\omega.$$

We want to take T out from the integral. Let $y = \omega T^r$, we obtain

$$u(T) = \frac{kT}{c^3} \int_0^\infty \frac{y^2}{T^{2r}} A(y)\, \frac{1}{T^r}\, dy = \frac{kT^{(1-3r)}}{c^3} \int_0^\infty y^2 A(y)\, dy.$$

Since the integral is constant, r must be -1 to satisfy Stefan's law (2). Therefore,

$$u_\omega = A(\omega/T)\, \frac{kT}{c^3}\, \omega^2 = \omega^3\, \frac{A(\omega/T)}{c^3}\, kT/\omega.$$

This is Wien's law. You can easily show that if u_ω has a maximum at a frequency, the frequency is proportional to T.

How do we construct a dimensionless quantity from ω/T? We know that the Boltzmann constant relates temperature to energy. Therefore it suggests a universal constant which relates frequency to energy. Planck's constant is indeed such a constant.

Acknowledgments

I acknowledge the use of ChatGPT to proofread and check the grammar of the text. This derivation is not my original. However, I lost the reference.

References

[1] G. H. Wannier, *Statistical Physics*. Dover, 1987.

[1] In case A's dependency is $\omega^s T^r$, we can always rewrite A as $A((\omega T^{r/s})^s)$, and replace A and r by $A'(x) = A(x^s)$ and $r' = r/s$.